

Interesting Tree (xortree2)

You are given a tree with N nodes, numbered from 1 to N . The edges are numbered from 1 to $N - 1$ and edge i has a weight w_i . The *difference* of two nodes u and v is the bitwise XOR of the edge weights on the simple path between u and v . Formally, if the simple path between u and v consists of edges $e_1, e_2, \dots, e_k$, then the difference of u and v is $w_{e_1} \oplus w_{e_2} \oplus \dots \oplus w_{e_k}$.

Let I be the set of *interesting* nodes (I is initially empty). Q operations are performed on this set: in one operation, a node is either added to or removed from I . After each operation, you have to find the maximum difference between two distinct interesting nodes.

Input

The first line contains two integers, N and Q . The i -th of the following $N - 1$ lines contains three integers u_i , v_i and w_i , corresponding to an edge of weight w_i between nodes u_i and v_i .

Each of the next Q lines contains a single integer p_i . If $p_i \notin I$, then node p_i gets added to I , otherwise it gets removed from I .

Output

You have to print Q lines. The i -th line must contain a single integer, the maximum difference between two distinct interesting nodes after the i -th operation. If the size of I is less than 2, print 0.

Constraints

- $1 \leq N \leq 50\,000$.
- $1 \leq Q \leq 50\,000$.
- $1 \leq u_i, v_i \leq N$ for each $i = 1 \dots N - 1$.
- $1 \leq p_i \leq N$ for each $i = 1 \dots Q$.
- $0 \leq w_i \leq 10^9$ for each $i = 1 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (21 points) $N, Q \leq 100$.

- **Subtask 3** (23 points) $w_i \in \{0, 1\}$ for each $i = 1 \dots N - 1$.

- **Subtask 4** (17 points) $N, Q \leq 1000$.

- **Subtask 5** (39 points) No additional limitations.

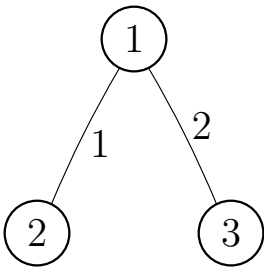

Examples

input	output
3 4 1 2 1 1 3 2 1 2 3 2	0 1 3 2
5 5 1 2 3 1 3 1 3 4 4 3 5 1 3 1 4 3 2	0 1 5 5 6

Explanation

In the **first sample case**, the set of interesting nodes and the interesting pair with the maximum difference after each operation:

Operation	Set of interesting nodes	Pair with maximum difference
1	{1}	No pairs
2	{1, 2}	(1, 2)
3	{1, 2, 3}	(2, 3)
4	{1, 3}	(1, 3)



In the **second sample case**, the set of interesting nodes and the interesting pair with the maximum difference after each operation:

Operation	Set of interesting nodes	Pair with maximum difference
1	{1}	No pairs
2	{1, 3}	(1, 3)
3	{1, 3, 4}	(1, 4)
4	{1, 4}	(1, 4)
5	{1, 2, 4}	(2, 4)

